

Technical Document #518: Computer Vision - Comprehensive Overview

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Pose estimation detects human body keypoints in images. It is used in action recognition, animation, and sports analysis.

Image generation creates new images from random noise or text descriptions. Diffusion models like Stable Diffusion produce photorealistic images.

Optical character recognition (OCR) converts images of text into machine-readable text. It is used in document digitization and automation.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Key Takeaways:

- Pose estimation detects human body keypoints in images.
- Image generation creates new images from random noise or text descriptions.
- Image classification assigns a label to an image from a fixed set of categories.